

# Evidence Helps on Average, but Sometimes Hurts: Characterizing Evidence-Induced Regressions in LLM Fact Verification

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Corrected submission draft for coauthor review (2026-09-28)

## Abstract

Supplying designated gold evidence is widely assumed to make language-model fact verification more reliable. We test that assumption in a paired experiment on 10,000 balanced FEVER Supported/Refuted claims, evaluating GPT-5.4 and GPT-5.4-mini without evidence and with FEVER-designated gold evidence (40,000 predictions). Evidence raises aggregate accuracy from 88.69% to 96.04% (+7.35 pp) for GPT-5.4 and from 84.67% to 95.54% (+10.87 pp) for GPT-5.4-mini. Those gains conceal 226 unique claims that were correct without evidence and become wrong after the same gold evidence is supplied (40 shared by both models, 186 specific to one). We describe these regressions with blinded five-judge progressive-disclosure adjudication by two model families, Grok and Claude, after repairing a structured-evidence stage whose historical inputs displayed corrupted sentence text. A descriptive taxonomy keeps judge decisiveness and agreement with FEVER separate. On the repaired inputs, the panels reach the FEVER label from the evidence sentences alone for 34.1% (Grok) and 46.9% (Claude) of regressions, reach a decisive verdict opposite to FEVER for 19.5% and 23.9%, and remain nondecisive for 31.4% and 18.1%. Claude is more decisive than Grok at every disclosure stage; titles and structured evidence resolve a minority of nondecisive cases and never reverse a decisive verdict; and exact per-claim category agreement between families is 76.1% ( $\kappa = 0.67$ ). The structured-stage analysis follows a dated amendment, written after judging, that raised the attempt limit and selected the accepted outputs; the results are stable when earlier valid outputs are substituted. Evidence can raise overall accuracy while creating localized failures, and how those failures distribute across adjudication outcomes depends on the adjudicating model family; the outcomes do not identify why GPT changed its answers.

*Keywords:* fact verification, FEVER, large language models, silver adjudication, LLM-as-judge, evidence-induced regressions

## 1 Introduction

Evidence grounding is a standard recipe for making language-model fact verification more trustworthy. If a model is shown the same designated evidence that a benchmark treats as sufficient, aggregate accuracy should rise, and remaining errors should look like residual retrieval or coverage

problems rather than failures to use gold evidence. That picture is incomplete. Aggregate improvements can hide a second phenomenon: claims the model already answered correctly without evidence, then answered incorrectly after the designated evidence was added.

This paper studies that phenomenon on FEVER [Thorne et al., 2018a]. We evaluate GPT-5.4 and GPT-5.4-mini on a balanced 10,000-claim Supported/Refuted sample under two paired conditions: claim only, and claim plus FEVER-designated gold evidence. Evidence helps in aggregate (Table 1). It also induces 226 unique correct-to-wrong regressions. Those regressions are rare relative to the 10,000-claim sample, but they are not unstructured noise. They concentrate on claim–evidence pairs that local NLI models treat as weakly warranted, and they are only partly shared across the two GPT models.

Counting regressions does not say what kind of claim–evidence pair produced them. An independent adjudicator may find the designated evidence decisive for the FEVER label, decisive for the opposite label, or insufficient; and its verdict may change when page titles or structured evidence are added. We therefore run blinded progressive-disclosure silver adjudication. Five isolated judges see sentence evidence (Stage A), then page titles (Stage B), then structured FEVER evidence with annotation boundaries and available alternative evidence (Stage C). A descriptive taxonomy records, separately, whether the panel is decisive at each stage, whether a decisive verdict agrees with FEVER, and at which stage a gold-agreeing verdict first appears. Two model families adjudicate the same 226 regressions: a locked Grok panel and a locked Claude panel.

An audit of the historical adjudication found that its Stage C packets displayed corrupted sentence text for most claims and that its three-way failure-mode taxonomy conflated judge–FEVER disagreement with evidence utilization. Under a post hoc correction protocol frozen before new judgments, Stage C was rebuilt from archival FEVER data and re-judged by both families; historical Stage A/B judgments, whose inputs were verified unchanged, are retained. Because provider capacity limits and concurrently running job runners produced more attempts than the protocol allowed, the analysis follows a dated post-judgment amendment, and we report how sensitive the results are to which valid output is used (Section 3.9).

The resulting claim is two-part. First, designated gold evidence improves overall accuracy while creating a measurable set of localized regressions. Second, independent judge panels describe those regressions as a mixture of FEVER-agreeing, FEVER-disagreeing and nondecisive outcomes whose proportions depend on the adjudicating family; the adjudication does not reveal why GPT changed its answer.

**Contributions.** We (i) quantify paired evidence gains and evidence-induced regressions on a verified 10,000-claim FEVER experiment; (ii) characterize regressions with confirmatory statistics, two local NLI diagnostics, and a 1,061-claim diagnostic cohort; (iii) describe the 226 unique regressions with blinded progressive-disclosure adjudication on repaired evidence, keeping decisiveness and FEVER agreement separate; (iv) compare two adjudicating families on the same claims and identify which descriptive findings depend on the family; and (v) document an evidence-reconstruction defect in the historical adjudication and its correction.

We ask four research questions, restated in Section 3.1. RQ1 asks whether designated gold evidence improves aggregate accuracy. RQ2 asks how frequent and cross-model-consistent the regressions are. RQ3 asks which observable adjudication outcomes characterize regressions under progressive disclosure. RQ4 asks which of those outcomes depend on the adjudicating model family.

## 2 Related Work

### 2.1 Evidence-based fact verification

Automated fact verification is commonly decomposed into claim detection, evidence retrieval, and verdict prediction [Guo et al., 2022]. FEVER [Thorne et al., 2018a] made the last two stages measurable at scale by pairing Wikipedia-derived claims with annotator-selected evidence sentences and ternary labels (Supported, Refuted, NotEnoughInfo). The first shared task already showed that retrieval and verification are both hard even when the evidence corpus is Wikipedia [Thorne et al., 2018b]. Later datasets extend the setting to scientific abstracts [Wadden et al., 2020], contrastive evidence revisions [Schuster et al., 2021], many-hop Wikipedia evidence [Jiang et al., 2020], and mixed sentence/table evidence [Aly et al., 2021]. Claim-only artifacts are also documented: models can exploit label cues that survive without evidence [Schuster et al., 2019]. Our experiment holds retrieval fixed: we supply FEVER-designated gold evidence and ask what happens when an LLM receives it. That removes open retrieval from the comparison, at the cost of studying a 2017-era, Wikipedia-centric, binary Supported/Refuted slice of the original task.

### 2.2 LLM reasoning with external evidence

LLMs do not treat supplied context as an oracle. Knowledge-conflict studies show that parametric memory can override or ignore contradictory context [Longpre et al., 2021]. Even when the relevant span is present, models are sensitive to where and how it is written [Liu et al., 2024] and can be pulled off-task by irrelevant material [Shi et al., 2023]. Wan et al. [Wan et al., 2024] show that models overweight surface relevance relative to credibility cues that humans treat as important. Mamta and Cocarascu [Mamta and Cocarascu, 2025] find that LLM fact verification on FEVER is brittle under small input perturbations. These results motivate a paired no-evidence versus gold-evidence design: if designated evidence still produces regressions, the failure is not missing retrieval.

### 2.3 Evidence sufficiency and ambiguity

Presence of evidence is not sufficiency. Atanasova et al. [Atanasova et al., 2022] show that fact-checking models often fail to detect when remaining evidence is insufficient. Glockner et al. [Glockner et al., 2024] document claim–evidence pairs that admit conflicting but valid interpretations and argue against forcing a single hard label. Our final-nondecisive outcome is a silver-adjudicator analogue of that problem: after full designated evidence, a panel still withholds Supported/Refuted. We do not treat that outcome, or a decisive verdict opposite to FEVER, as proof that a FEVER label is wrong.

### 2.4 LLM-as-judge and evaluator sensitivity

Using LLMs as judges scales evaluation but imports judge-specific biases, including position, verbosity, and self-preference effects [Zheng et al., 2023, Liu et al., 2023, Chen et al., 2024]. Position bias itself varies by judge and task rather than appearing as a uniform artifact [Shi et al., 2025]. Huang et al. [Huang et al., 2025] show that judge models are not interchangeable substitutes for one another. We therefore treat judge-family disagreement as a result. Grok and Claude are two

silver instruments applied to the same blinded packets. Where they agree, a descriptive finding does not hinge on one family; where they differ, no single family’s category shares can be taken as the prevalence. Neither panel is human ground truth.

### 3 Study Design

#### 3.1 Research questions

**RQ1** How does providing FEVER-designated gold evidence affect aggregate LLM fact-verification accuracy?

**RQ2** When evidence changes a previously correct prediction into an incorrect one, how frequent and consistent are those regressions across models?

**RQ3** Which observable adjudication outcomes—decisiveness, agreement with FEVER, and the disclosure stage at which they change—characterize evidence-induced regressions under progressive disclosure?

**RQ4** Which of those outcomes depend on the adjudicating model family?

#### 3.2 FEVER dataset and sampling

We sample from the FEVER development set [Thorne et al., 2018a]. The manuscript experiment is a balanced 10,000-claim extract: 5,000 Supported and 5,000 Refuted, seed 42, **NOT ENOUGH INFO** excluded. Gold evidence is resolved from official FEVER Wikipedia shards into the text shown to the models. Sampling and validation artifacts are committed with the frozen repository; third-party FEVER and Wikipedia text are not relicensed by the project MIT license.

#### 3.3 Models and paired evidence conditions

Each claim is evaluated four times: GPT-5.4 and GPT-5.4-mini, each under `claim_only` and `claim_plus_evidence`. That yields 40,000 predictions. The evidence condition uses the designated FEVER evidence, not an open-web retrieval step. Original GPT runs stored labels only, so later analysis cannot recover confidence, rationales, or token-level attributions.

#### 3.4 Transition definitions

For each model, a claim has one of four paired transitions: *robust* (correct without and with evidence), *rescue* (wrong without, correct with), *resistant* (wrong in both), or *regression* (correct without, wrong with). A unique-claim regression is a claim that regresses for at least one of the two GPT models. Unique-claim counts are never treated as model-observations: 115 GPT-5.4 regressions and 151 mini regressions are 226 unique claims, not 266.

#### 3.5 Analysis v2

Analysis v2 treats the 40,000-row experiment file as immutable source data. It rebuilds the paired dataset, runs McNemar tests and bootstrap confidence intervals, extracts linguistic and evidence-

structure features, embeds claim–evidence pairs, and applies two local NLI models (nli-deberta-v3-small and distilbert-base-uncased-mnli) as diagnostics, not as ground truth. Leakage audits and an independent 69-check headline verifier are part of the frozen record. Confirmatory tests were pre-specified; NLI and feature analyses are exploratory.

### 3.6 Diagnostic cohort

The adjudication cohort contains 1,061 unique claims: all 226 unique regressions, 335 resistant, 250 rescue, and 250 robust. One item (SA-000352) is provider-filtered, so judged  $N = 1,060$ . The Grok panel judges the 1,060-claim cohort; the Claude panel judges only the 226 regressions. This diagnostic sample is not representative of all 10,000 claims, and cross-family comparisons always pair the same 226 claims.

### 3.7 Progressive-disclosure adjudication

Five isolated judges per stage and batch see: Stage A, the original selected gold-evidence sentence text (identical to the GPT evidence prompt); Stage B, Stage A plus the page titles; Stage C, structured FEVER evidence—ordered sentence pointers, annotation boundaries, and the text of available alternative evidence sets. Stage C can therefore add information as well as change representation: on the repaired inputs it adds sentence text beyond the selected sentences for 149 cohort claims, including 41 regressions. Judges do not browse, retrieve new pages, or see FEVER labels, GPT predictions, cohort tags, NLI scores, or other judges. Each stage is frozen before the next stage’s packets are built, and packets carry only current-stage evidence. Consensus is computed by code from the five raw judgments; resolver outputs never replace raw consensus. The Grok panel is locked to `cursor-grok-4.6-high-fast` and the Claude panel to `claude-opus-5-thinking-high`.

### 3.8 Descriptive taxonomy

For each claim the taxonomy records decisiveness (Supported or Refuted versus Ambiguous or Unresolved) and FEVER agreement at every stage, the first gold-agreeing stage, title and Stage C resolutions, reversals, and losses of decisiveness. Each claim then receives exactly one category, in priority order: *final nondecisive* (Stage C consensus Ambiguous or Unresolved); *decisive judge–FEVER disagreement* (decisive Stage C verdict opposite to FEVER); *nonmonotonic judge path* (final gold agreement after a reversal or loss of decisiveness); *sentence-only gold-agreeing* (gold-agreeing from Stage A onward); *title-disclosure gold-agreeing* (first gold agreement at Stage B); and, for first gold agreement at Stage C, *additional-evidence gold-agreeing* when Stage C adds sentence text and *structured-disclosure gold-agreeing* otherwise. These categories describe where a blinded panel’s verdict stands; they are not causal mechanisms inside GPT. A decisive disagreement with FEVER is not evidence that the FEVER label is wrong, and sentence-only gold agreement is compatible with, but does not establish, a failure by GPT to use the supplied evidence. The taxonomy was specified post hoc, after inspection of the historical results, and awaits coauthor review.

### 3.9 Evidence audit and Stage C correction

The historical Stage C packets took each evidence line’s last tab-separated field, often a hyperlink target, instead of the sentence text. Chosen-set text differed from the canonical sentences for 936 of 1,061 cohort claims (190 of the 226 regressions); ten further claims lacked structured text because decomposed-Unicode titles were missing from a cache. Exact selected evidence sets were recovered by recomputing the original set identifiers, and all selected and alternative sentences were rebuilt from local archival FEVER data. Stage A inputs equal the GPT evidence prompts and Stage B inputs equal the historical records exactly; their frozen judgments are retained under the documented fresh-context-per-stage protocol, and all Stage C judgments are new. Consensus paths therefore combine historical Stage A/B judgments with new Stage C judgments.

A post hoc correction protocol, frozen on 2026-09-27 before any new judgment, fixed the repaired packets, the taxonomy, the statistical analyses, and execution rules: at most three attempts per job, acceptance of the first schema-valid response, and a distinct session per attempt. Seventy required jobs (55 Grok, 15 Claude; 6,430 new Stage C judgments) all delivered schema-valid outputs.

**Post-judgment amendment.** Provider quota and account usage limits, together with job runners that ran concurrently for part of the execution, left six jobs with four to seven attempt records and 13 failed-attempt records whose session is marked unknown. The runners never read model outputs; retries depended only on exit status. After all outputs existed and had been inspected, we adopted a dated amendment (2026-09-28) that changes three execution rules and nothing else: (i) up to nine attempts per job; (ii) a record with an unknown session counts as a launched attempt if the agent event log that survives in its attempt slot identifies exactly one agent used nowhere else (all 13 records met this test, and all 101 launched sessions are distinct); and (iii) each job’s analysed output is the output its execution procedure accepted. For three jobs that output is not the first schema-valid one: in two Claude jobs earlier finished runs were not accepted because a runner collision had occupied their slots, and in one Grok job a run wrote a complete output before ending in a provider error. We therefore repeat every analysis with those first valid outputs substituted (Appendix A). The amendment is not preregistered; under the original rules the protocol’s primary analysis remains incomplete. Packets, models, taxonomy, statistics and analysis code are unchanged, and the protocol’s follow-up panels (a newly selected residual Claude panel and a Stage C resolver) were not run.

### 3.10 Supporting cross-family data

GLM-5.3 contributes a completed Stage A panel compared to frozen Cursor Stage A ( $n = 1,060$ ); Stage A inputs are unaffected by the audit. A historical Claude Stage C test on 231 Grok leftovers used the defective Stage C packets and a residual set selected with defective Grok Stage C judgments; it is reported only as a historical observation (Appendix C).

### 3.11 Statistical analysis

Paired accuracy uses McNemar tests; regression and rescue rates use bootstrap confidence intervals. Descriptive adjudication rates use Wilson 95% intervals. Cross-family comparisons on the paired 226 claims use exact McNemar tests with Benjamini–Hochberg correction across 14 tests; shared-versus-model-specific nondecisiveness uses odds ratios in a separate two-test family. Cross-family

**Table 1:** Paired behavioral results on 10,000 balanced FEVER Supported/Refuted claims (40,000 predictions). Accuracies and transition counts are from the verified Analysis v2 tables. Unique-claim regression counts treat a claim as one unit even if both models regress.

| Model                                                                              | Claim-only | +Gold ev. | $\Delta$ (pp) | Regressions | Rescues       |
|------------------------------------------------------------------------------------|------------|-----------|---------------|-------------|---------------|
| GPT-5.4                                                                            | 88.69%     | 96.04%    | +7.35         | 115 (1.15%) | 850 (8.5%)    |
| GPT-5.4-mini                                                                       | 84.67%     | 95.54%    | +10.87        | 151 (1.51%) | 1,238 (12.4%) |
| Unique claims that regress for at least one model: 226 (40 both; 186 exactly one). |            |           |               |             |               |

**Table 2:** Adjudication outcomes for the 226 unique regressions on repaired Stage C inputs, as counts with percentages of 226.  $\Delta$  is Claude minus Grok in percentage points with exact McNemar tests, Benjamini–Hochberg adjusted over 14 cross-family tests. “FV” gives the Claude count under the first-valid sensitivity selection (= unchanged); Grok regression-cohort values are identical under both selections. Categories describe blinded judge-panel verdicts; they are not causes of GPT errors or corrected FEVER labels.

| Category                                                                                                                               | Grok       | Claude      | FV  | $\Delta$ (pp) | $p_{BH}$             |
|----------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|-----|---------------|----------------------|
| Sentence-only gold-agreeing                                                                                                            | 77 (34.1%) | 106 (46.9%) | 105 | +12.8         | $4.6 \times 10^{-7}$ |
| Title-disclosure gold-agreeing                                                                                                         | 10 (4.4%)  | 3 (1.3%)    | =   | −3.1          | 0.13                 |
| Structured-disclosure gold-agreeing                                                                                                    | 2 (0.9%)   | 3 (1.3%)    | 6   | +0.4          | 1.00                 |
| Additional-evidence gold-agreeing                                                                                                      | 17 (7.5%)  | 14 (6.2%)   | =   | −1.3          | 0.58                 |
| Nonmonotonic judge path                                                                                                                | 5 (2.2%)   | 5 (2.2%)    | =   | 0.0           | 1.00                 |
| Decisive judge–FEVER disagreement                                                                                                      | 44 (19.5%) | 54 (23.9%)  | =   | +4.4          | 0.015                |
| Final nondecisive                                                                                                                      | 71 (31.4%) | 41 (18.1%)  | 39  | −13.3         | $7.2 \times 10^{-8}$ |
| Exact per-claim category agreement: 172/226, 76.1% (Wilson 70.1–81.2), $\kappa = 0.673$ ; first-valid 168/226, 74.3%, $\kappa = 0.651$ |            |             |     |               |                      |

taxonomy agreement is reported as exact match and Cohen’s  $\kappa$ . All adjudication analyses are post hoc.

## 4 Results

### 4.1 Gold evidence improves aggregate accuracy

Table 1 answers RQ1. On the same 10,000 claims, designated gold evidence raises GPT-5.4 accuracy from 88.69% to 96.04% (+7.35 pp) and GPT-5.4-mini from 84.67% to 95.54% (+10.87 pp). Paired McNemar tests are large (GPT-5.4  $\chi^2 = 558.3$ ,  $p \approx 2 \times 10^{-123}$ ; mini  $\chi^2 = 849.1$ ,  $p \approx 10^{-186}$ ). Figure 1 shows the four-way transition structure behind those averages. Evidence also increases cross-model prediction agreement, from 89.3% ( $\kappa = 0.786$ ) without evidence to 97.2% ( $\kappa = 0.944$ ) with evidence.

### 4.2 Evidence can induce correct-to-wrong regressions

The same paired design produces 115 GPT-5.4 regressions (1.15%, bootstrap 95% CI [0.95, 1.36]) and 151 mini regressions (1.51%, CI [1.28, 1.75]). Rescues are more common: 850 (8.5%) and 1,238 (12.4%). Among claim-only errors, evidence corrects 75.2% (GPT-5.4) and 80.8% (mini).

Unique-claim counts, not model-observations, are the regression cohort for later adjudication: 226 claims regress for at least one model, 40 for both, and 186 for exactly one (75 GPT-5.4-only,

**Table 3:** Stage-wise nondecisiveness and resolution for the 226 regressions (percentages of 226). Stage A and B verdicts are retained historical judgments on unchanged inputs; Stage C verdicts are new judgments on repaired inputs.  $\Delta$  is Claude minus Grok with exact McNemar tests, Benjamini–Hochberg adjusted over 14 tests. Brackets give the first-valid sensitivity selection where it differs.

| Measure                   | Grok        | Claude          | $\Delta$ (pp) | $p_{BH}$                                      |
|---------------------------|-------------|-----------------|---------------|-----------------------------------------------|
| Nondecisive at Stage A    | 103 (45.6%) | 66 (29.2%)      | −16.4         | $1.1 \times 10^{-8}$                          |
| Nondecisive at Stage B    | 95 (42.0%)  | 64 (28.3%)      | −13.7         | $9.4 \times 10^{-7}$ [ $1.2 \times 10^{-6}$ ] |
| Nondecisive at Stage C    | 71 (31.4%)  | 41 (18.1%) [39] | −13.3 [−14.2] | $7.2 \times 10^{-8}$ [ $9.1 \times 10^{-8}$ ] |
| Resolved by titles (A→B)  | 16 (7.1%)   | 9 (4.0%)        | −3.1          | 0.25                                          |
| Resolved at Stage C (B→C) | 27 (11.9%)  | 24 (10.6%) [27] | −1.3 [0.0]    | 0.93 [1.00]                                   |
| Decisive verdict reversed | 0           | 0               | 0.0           | 1.0                                           |

**Table 4:** Final Stage C nondecisiveness among regressions shared by both GPT models versus regressions specific to one model. Odds ratios compare shared with model-specific regressions (two-test Benjamini–Hochberg family). Brackets give the Claude first-valid sensitivity selection.

| Subgroup                  | $n$ | Grok final nondecisive | Claude final nondecisive |
|---------------------------|-----|------------------------|--------------------------|
| Shared by both GPT models | 40  | 18 (45.0%)             | 14 (35.0%)               |
| Exactly one GPT model     | 186 | 53 (28.5%)             | 27 (14.5%) [25 (13.4%)]  |
| Odds ratio (95% CI)       |     | 2.05 (1.02–4.13)       | 3.17 (1.47–6.83) [3.47]  |
| $p_{BH}$                  |     | 0.059                  | 0.011 [0.0046]           |

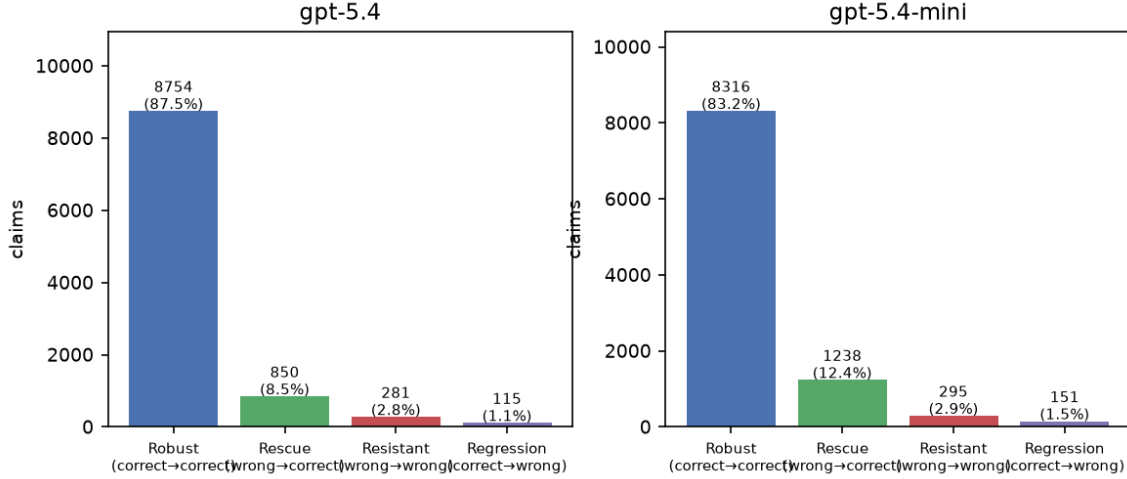
111 mini-only). A paired McNemar test on discordant claims finds a modest model difference in regression incidence (115 vs. 151,  $\chi^2 = 6.59$ ,  $p = 0.010$ ). 87.3% of claims share the same transition class across models. RQ2’s answer is therefore: regressions are real, rare relative to 10,000 claims, only partly shared, and large enough to study as a structured minority. These behavioral results come from the original 40,000 predictions and are unaffected by the adjudication audit; they are observed outcomes of the original prompts, not evidence of repeat-run stability.

### 4.3 Characteristics of evidence-induced regressions

Regressions are not a random slice of the sample. A local NLI diagnostic (nli-deberta-v3-small) disagrees with the FEVER label on 74.8% of GPT-5.4 regressions and 76.8% of mini regressions, against 22.6% of GPT-5.4 robust successes (Figure 2). Resistant failures look similar (75.8%). A second, architecturally different NLI model reproduces the same ordering. The 2,462 claims flagged as weakly warranted also have lower claim-only GPT-5.4 accuracy (81.0% vs. 88.69%), so the NLI signal marks hard items rather than a quirk of one evidence-condition model.

Label asymmetry is model-dependent. GPT-5.4 regressions concentrate on Supported claims (88 vs. 27 Refuted,  $p = 1.8 \times 10^{-8}$ ); mini shows no such split (74 vs. 77,  $p = 0.87$ ). Surface features (length, overlap, evidence-set size) only weakly separate regressions. Predictive models that include NLI features do better than surface-only models, but rare-class PR-AUC remains low. We treat this block as characterization; it does not depend on the adjudication stage that was corrected.

Q1: Four-way evidence transition classes per model (n=10,000 each)



**Figure 1:** Paired transition matrices for GPT-5.4 and GPT-5.4-mini. Off-diagonal cells are rescues (wrong→correct) and regressions (correct→wrong).

#### 4.4 Adjudication outcomes on repaired evidence

Table 2 and Figure 3 answer RQ3 for the 226 regressions. Values in this and the next subsection use the accepted Stage C outputs; the first-valid sensitivity selection is given where it differs (Section 3.9). The largest Grok categories are sentence-only gold agreement (77; 34.1%, Wilson 95% CI 28.2–40.5) and final nondecisiveness (71; 31.4%, 25.7–37.7), whose intervals overlap. Decisive verdicts that disagree with FEVER form the third group (44; 19.5%, 14.8–25.1). Title, structured and additional-evidence gold agreement together cover 29 claims (12.8%), and five claims follow a nonmonotonic path. Among regressions with a decisive Stage C consensus, the Grok verdict agrees with FEVER on 111 of 155 (71.6%).

Progressive disclosure changes a minority of verdicts (Table 3, Figure 4). Titles make 16 of the 103 Stage A-nondecisive regressions decisive, and Stage C makes 27 of the 95 Stage B-nondecisive regressions decisive. No decisive consensus is reversed by titles or by Stage C. Because Stage C adds alternative sentence text for 41 regressions, Stage C-associated changes cannot be attributed to representation alone; the 17 additional-evidence cases are separated from the two structure-only cases for that reason.

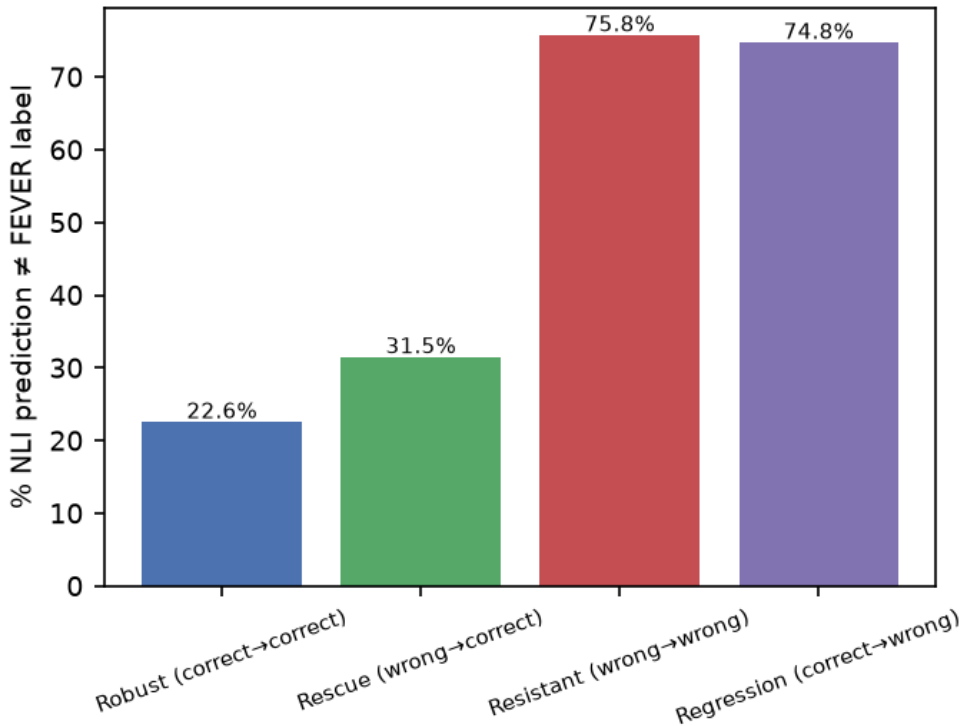
Shared regressions remain nondecisive more often than model-specific ones (Table 4): 45.0% versus 28.5% under Grok (odds ratio 2.05,  $p_{BH} = 0.059$ ).

The RQ3 answer is a mixture of FEVER-agreeing, FEVER-disagreeing and nondecisive outcomes, with no single dominant category under Grok. These outcomes do not show why GPT changed its answer, and the decisive disagreements are not evidence that FEVER is wrong.

#### 4.5 Cross-family comparison

The Claude panel judges the same 226 regressions (1,130 new Stage C judgments, paired with its 2,260 retained Stage A/B judgments). Claude assigns more regressions to sentence-only gold agreement (106, 46.9%, CI 40.5–53.4; first-valid 105; +12.8 and +12.4 pp versus Grok,  $p_{BH} \leq$

QE: Local NLI / FEVER disagreement by transition class (gpt-5.4)



**Figure 2:** Local NLI disagreement with the FEVER label by GPT-5.4 transition class. The NLI models are diagnostics, not a relabeling of FEVER.

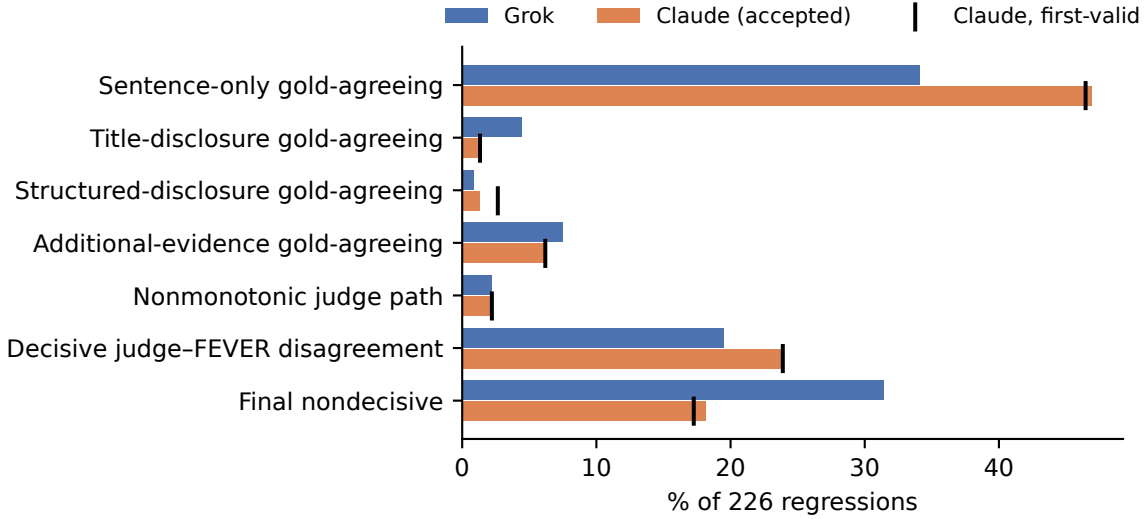
$2.1 \times 10^{-6}$ ) and fewer to final nondecisiveness (41, 18.1%, CI 13.7–23.7; first-valid 39;  $-13.3$  and  $-14.2$  pp,  $p_{\text{BH}} \leq 9.1 \times 10^{-8}$ ). Decisive judge–FEVER disagreement is somewhat more common under Claude (54 versus 44;  $+4.4$  pp,  $p_{\text{BH}} = 0.015$ ). Among decisive Stage C consensus verdicts, FEVER agreement is similar across families (Claude 131/185, 70.8%; Grok 111/155, 71.6%).

Claude is more decisive at every stage (Table 3, Figure 4): nondecisive rates are 29.2%, 28.3% and 18.1% (first-valid 17.3%) versus Grok’s 45.6%, 42.0% and 31.4%. Titles resolve 9 of Claude’s 66 Stage A-nondecisive regressions and Stage C resolves 24 of its 64 Stage B-nondecisive regressions (first-valid 27). Title and Stage C resolution rates do not differ significantly between families, and neither family reverses a decisive consensus.

Under Claude, shared regressions are also more often nondecisive than model-specific ones (35.0% versus 14.5%; odds ratio 3.17,  $p_{\text{BH}} = 0.011$ ; first-valid 3.47,  $p_{\text{BH}} = 0.005$ ).

Exact per-claim category agreement is 172/226 (76.1%, Wilson 70.1–81.2;  $\kappa = 0.673$ ) for the accepted selection and 168/226 (74.3%, 68.3–79.6;  $\kappa = 0.651$ ) for first-valid. The first-valid figure coincides numerically with the historical 74.3%, which measured a different, withdrawn taxonomy on defective inputs; the two are not comparable.

**Selection sensitivity.** Substituting the first valid outputs for the accepted ones changes Stage C consensus for six of the 226 Claude claims and for one of the 1,060 Grok claims (a rescue-cohort claim, so no Grok regression result changes). All 14 cross-family tests and both odds-ratio tests keep their Benjamini–Hochberg decisions, and the only sign change is the non-significant Stage C



**Figure 3:** Adjudication outcome categories for the 226 unique regressions on repaired Stage C inputs, by adjudicating family. Bars use the accepted outputs; black ticks mark Claude under the first-valid sensitivity selection. Grok values are identical under both selections.

resolution difference ( $-1.3$  to  $0.0$  pp). Across all 12 combinations of completed runs for the two affected Claude jobs, zero to six claims change consensus relative to the accepted outputs and Claude Stage C Ambiguous counts range from 37 to 42 of 226.

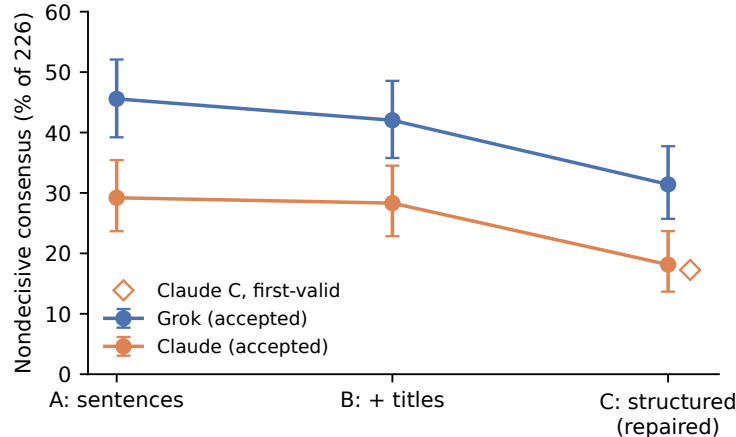
#### 4.6 Supporting Stage A check

GLM-5.3 Stage A, compared with frozen Cursor Stage A on 1,060 items, yields modal verdict agreement 0.862 and consensus agreement 0.831. GLM is less ambiguity-conservative (ambiguous/unresolved rate 0.223 vs. Cursor 0.291). This is Stage A robustness only; Stage A inputs were unaffected by the audit.

#### 4.7 What depends on the judge family

Common to both families: all categories occur; each disclosure step resolves a minority of the regressions that were nondecisive before it and never reverses a decisive consensus; roughly a fifth to a quarter of regressions end in a decisive verdict opposite to FEVER; decisive verdicts agree with FEVER at similar rates; and shared regressions are more often nondecisive (significant under Claude only).

Family-dependent: absolute nondecisiveness at every stage (13–16 pp lower under Claude), the share of sentence-only gold agreement, and therefore which category is largest. Per-claim assignment agrees on roughly three claims in four ( $\kappa = 0.65$ – $0.67$ ). RQ4’s answer is that the qualitative mixture is shared while its proportions are family-dependent. The design cannot decide whether Claude is better calibrated or more overconfident on thin warrants, because both panels are models scoring the same packets.



**Figure 4:** Nondecisive (Ambiguous or Unresolved) five-judge consensus on the 226 regressions at each disclosure stage, with Wilson 95% intervals. Stages A and B are retained historical judgments on unchanged inputs; Stage C is new judgments on repaired inputs. The open diamond is Claude Stage C under the first-valid sensitivity selection.

## 5 Discussion

### 5.1 Evidence helps globally but can hurt locally

The practical headline is not that gold evidence fails. It succeeds on average, and it rescues far more claim-only errors than it creates. The scientific headline is that those averages are the wrong unit for reliability engineering. A 1.15–1.51% regression rate is small in a 10,000-claim table and large if each regression is a previously correct decision that evidence reversed. Evaluation reports that publish only condition-wise accuracy will miss that reversal.

### 5.2 What adjudication can and cannot say about regressions

For roughly a third (Grok) to nearly half (Claude) of the 226 regressions, a blinded panel reaches the FEVER label from the evidence sentences alone and keeps it as titles and structured evidence are added. On those items the designated evidence appears adequate to an independent reader while GPT moved away from the correct label after receiving it. That pattern is compatible with an evidence-utilization failure and connects to work showing that models can ignore, mishandle, or be distracted by supplied context [Longpre et al., 2021, Shi et al., 2023, Wan et al., 2024]. It does not establish that explanation: the original runs stored labels only, and the adjudicator is a different model reading the evidence in a different prompt. A further fifth to quarter of regressions end in a decisive verdict opposite to FEVER, and a sixth (Claude) to nearly a third (Grok) end nondecisive. In neither case does a decisive GPT answer after evidence show that GPT failed to use evidence supporting the FEVER answer. These observations therefore do not establish the prevalence or cause of GPT evidence-utilization failures.

### 5.3 Nondecisiveness and disagreement with FEVER

Final nondecisiveness is common in both families, which is expected given prior evidence that sufficiency and ambiguity are first-class problems in fact verification [Atanasova et al., 2022, Glockner et al., 2024]. We do not interpret nondecisive outcomes or decisive judge–FEVER disagreements as FEVER annotation errors; inspected rationales show substantive conflicts (for example, a date or scope mismatch between claim and evidence), but judge rationales are observations, not labels. Shared regressions, which both GPT models got wrong after evidence, are more often nondecisive under both families, although the Grok difference is not significant after correction.

### 5.4 Disclosure and representation

Titles and structured evidence change a minority of verdicts and never reverse a decisive consensus. Stage C adds alternative sentence text for some claims, so its effect mixes added information with changed representation. Separating the two would require a structure-only arm that shows the chosen evidence set without alternatives; we did not run one. We therefore make no claim about the size or portability of a pure representation effect.

### 5.5 Judge-family dependence as measurement uncertainty

The Claude-versus-Grok difference is a result, not a defect to be explained away. LLM-as-judge work already shows that evaluators have family-specific biases and are not drop-in substitutes [Zheng et al., 2023, Liu et al., 2023, Chen et al., 2024, Huang et al., 2025]. Here the same packets, rubric, and taxonomy produce a 13–16 pp difference in nondecisiveness at every stage and disagreement on about one regression in four. Category shares are therefore properties of an adjudicating family as much as of the claims; two instruments do not define an uncertainty interval for a latent prevalence, and we report both rather than a single estimate.

### 5.6 Implications for evidence-grounded LLM evaluation

Four implications follow. First, paired no-evidence versus gold-evidence evaluation should be standard whenever a benchmark supplies designated evidence; otherwise regressions are invisible. Second, automated diagnostic categories need a second judge family before their shares are treated as stable, and decisiveness should be reported separately from agreement with the benchmark label. Third, residual nondecisiveness should remain a first-class output of adjudication rather than being forced into Supported/Refuted. Fourth, adjudication inputs deserve the same verification as model inputs: a field-extraction error silently corrupted the structured stage of our original adjudication. Human review of the 54–58 claim-level Grok–Claude category disagreements would be the natural next measurement, not another automated family.

## 6 Limitations

All adjudication is automated silver labeling. Grok, GLM, and Claude judgments are not expert human annotation and do not create a new gold standard. Within-family agreement overstates how much independent human raters would agree, because the five judges are isolated contexts of one locked model.

Category shares are judge-family-sensitive. We report both complete panels and refuse a single point estimate.

FEVER labels remain the behavioral gold standard. NLI disagreement and judge-FEVER disagreement mark candidate ambiguity; they do not prove an annotation is wrong.

The study is binary Supported/Refuted FEVER, 2017-era Wikipedia claims, and designated gold evidence rather than retrieved open-web context. `NOT ENOUGH INFO` is excluded by design.

Both evaluated systems are GPT-5.4 variants. Transition concordance can reflect family-correlated errors.

Progressive disclosure locates where a judge panel’s verdict changes. It does not observe GPT-internal causal mechanisms, and the original GPT runs stored labels only.

The historical adjudication’s Stage C packets displayed corrupted sentence text, and its taxonomy conflated disagreement with FEVER with evidence utilization. The repaired Stage C and the descriptive taxonomy were specified post hoc, after the historical results and defects were inspected; the correction protocol is not a preregistration and the taxonomy awaits coauthor review. Stage C adds alternative evidence text for some claims, so Stage C-associated changes are not a pure representation manipulation.

Consensus paths combine retained historical Stage A/B judgments with new Stage C judgments. The historical protocol specified a fresh context per judge, batch and stage, but raw historical session transcripts were not available to verify runtime isolation independently.

The Stage C analysis rests on an amendment adopted after the judgments and their provisional results had been seen: it raised the attempt limit from three to nine, accepted unknown-session records on the strength of surviving agent logs, and selected the accepted outputs where earlier valid outputs existed. Under the original rules the protocol’s primary analysis is incomplete, and the amended analysis is not a pre-specified result. The amendment changed no estimate, and the first-valid alternative moves at most six Claude claims without changing any significance decision; but provenance still rests on operator records and hashed logs rather than provider attestation, and one runner collision record was deleted during execution.

The protocol’s follow-up panels were not run: a newly selected residual Claude panel, a Stage C resolver, and an optional fixed historical residual panel. The Claude panel covers the 226 regressions, not the 1,060-item judged cohort, and GLM is Stage A only. Historical execution events in the original Claude panel (a quarantined path-leak rerun, a usage-limit pause, and a duplicate-batch tie-break) affect only its retained Stage A/B judgments and are documented in Appendix B.

## 7 Conclusion

Designated gold evidence improves GPT-5.4 and GPT-5.4-mini fact verification in aggregate and still induces 226 unique correct-to-wrong regressions. These behavioral results are unchanged by the adjudication audit. Re-adjudicated on repaired structured evidence by two model families, the regressions form a mixture of FEVER-agreeing, FEVER-disagreeing and nondecisive panel outcomes; disclosure of titles and structured evidence resolves a minority of the nondecisive cases and never reverses a decisive verdict. The mixture is common to both families, but its proportions are not: Claude is 13–16 pp more decisive at every stage and places more regressions in sentence-only gold agreement, and the two families assign the same category to about three regressions in four. The historical three-way failure-mode shares and their cross-family ordering are withdrawn,

and the adjudication does not identify why GPT changed its answers. Paired no-evidence versus gold-evidence evaluation, a second judge family, and verified adjudication inputs are what it takes to see these failures and describe them without overstating them.

## Data and code

The behavioral record is tag `evidex-artifact-v1` at commit `29102c3`; its headline verifiers and test suites are unchanged. The evidence audit, repaired Stage C packets, frozen correction protocol (2026-09-27), dated Stage C amendment (2026-09-28), all execution records including failed attempts, and the amended and sensitivity analyses are on branch `codex/methodology-correction-v2`. Project code is MIT-licensed; FEVER and Wikipedia-derived files are third-party material.

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## A Stage C amendment and selection sensitivity

The correction protocol (frozen 2026-09-27, before any new judgment) allowed three attempts per job, required the first schema-valid response to be accepted, and required a distinct session per attempt. The dated amendment (2026-09-28), written after all 70 outputs existed and had been inspected, replaces these with: up to nine attempts; an unknown-session record counts as a launched attempt when the agent event log surviving in its slot names exactly one agent used in no other slot; and the accepted output of each job is analysed. Every other provenance check (locked model routing, fresh context, input and output hashes, raw output identical to the delivered output, and exactly one accepted attempt that is the last attempt) is applied unchanged, as are the packets, taxonomy, statistics and analysis code. Under the amendment all 70 jobs pass, and the amended analysis gives exactly the estimates of the provisional analysis computed before the amendment was written.

Three jobs have an earlier schema-valid output than the accepted one. In two Claude jobs, two and three earlier runs finished but were not accepted because a concurrent runner had already written a failure record into their attempt slots. In one Grok job, the first run wrote a complete output and then ended in a provider error. The first-valid sensitivity selection substitutes the earliest of these outputs in all three jobs.

| Quantity (of 226 regressions)                     | Accepted                      | First-valid                   |
|---------------------------------------------------|-------------------------------|-------------------------------|
| Claude Stage C nondecisive                        | 41 (18.1%)                    | 39 (17.3%)                    |
| Claude sentence-only gold-agreeing                | 106                           | 105                           |
| Claude structured-disclosure gold-agreeing        | 3                             | 6                             |
| Claude Stage C consensus labels changed           | —                             | 6                             |
| Grok Stage C labels changed (of 1,060 claims)     | —                             | 1, not a regression           |
| Exact Grok–Claude category agreement              | 172 (76.1%), $\kappa = 0.673$ | 168 (74.3%), $\kappa = 0.651$ |
| Claude shared-vs-specific odds ratio ( $p_{BH}$ ) | 3.17 (0.011)                  | 3.47 (0.0046)                 |
| Benjamini–Hochberg decisions changed (16 tests)   | —                             | 0                             |

Across all 12 combinations of the completed runs for the two Claude jobs, Stage C consensus differs from the accepted selection on zero to six claims, and the Claude Stage C Ambiguous count ranges from 37 to 42.

## B Historical execution events in the Claude full panel

The historical Claude full panel reused the locked model `claude-opus-5-thinking-high`, the shared rubric, and the blinded Stage A/B/C evidence files; its Stage A/B judgments are retained in this draft and its historical Stage C judgments are not used. During the first Stage A batch-01 wave, judge-facing file paths included the panel directory name and therefore the cohort-selection criterion. Before inspecting the affected judgments, the five delivered files ( $5 \times 77 = 385$  valid records) were quarantined, judge I/O moved to a neutral directory, and that batch rerun under the unchanged model, rubric, and evidence packets. No quarantined verdict was read. Stage B was interrupted by a usage limit after 601 of 1,130 judgments; work resumed on the same Claude slug and already-valid judgments were kept. Recovery overlap caused three judge-batches to be produced twice; a first-ingested-wins rule, applied without reading either copy’s labels, retained one file per (stage, judge, item).

## C Withdrawn historical diagnostic results

The following results appeared in the historical draft ([paper/](#), preserved unchanged). They were computed with the legacy three-way taxonomy on Stage C packets that displayed corrupted sentence text, and they are withdrawn as estimates. They are listed only so readers can trace the correction.

- Grok three-way split of the 226 regressions: 50.0% evidence-utilization failure, 11.9% representation-sensitive, 38.1% residual ambiguity (113/27/86). Of the 113 legacy utilization cases, 36 had a decisive Stage C verdict opposite to FEVER.
- Claude three-way split: 65.9% / 5.3% / 28.8% (149/12/65); 46 of the 149 legacy utilization cases disagreed with FEVER at Stage C.
- “Utilization failure is largest in both panels” and the claimed replication of the category ordering; the 50–66% range for utilization failure.
- Legacy taxonomy agreement 74.3% (168/226),  $\kappa = 0.537$ , and the four-item representation-sensitive overlap.
- Historical Stage C nondecisive rates (Grok 38.1%, Claude 28.8%) and the Stage C-dependent McNemar test.
- The historical Claude residual test (134/231, 58.0%, persistently ambiguous), whose inputs and residual selection both depended on defective Stage C.

Historical figures of the category split, taxonomy confusion, Stage C ambiguity and Grok resolution flow are not reproduced in this draft.

## D Denominator checklist

| Quantity                                         | $n$    |
|--------------------------------------------------|--------|
| FEVER claims in the paired experiment            | 10,000 |
| GPT predictions                                  | 40,000 |
| Diagnostic-cohort claims constructed             | 1,061  |
| Judged diagnostic-cohort items                   | 1,060  |
| Unique regressions                               | 226    |
| New Stage C judgments, Grok ( $1,060 \times 5$ ) | 5,300  |
| New Stage C judgments, Claude ( $226 \times 5$ ) | 1,130  |
| Required correction jobs delivered               | 70     |
| Retained historical Claude Stage A/B judgments   | 2,260  |
| Optional historical-residual judgments (not run) | 1,155  |